

A Steganographic Approach For Medical Imaging: Enhancing Image Quality And Security

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Abstract—In this project plan, we present the study of steganography applied to medical imaging, its different methodologies, associated challenges, limitations, and potential solutions. Steganographic techniques such as LSB, SVM and DWT methods, are analysed for securing patient sensitive data in medical images. Challenges include balancing image resolution, contrast, vulnerability to attacks, and potential impacts on patient records. Steganography conceals sensitive data within images, ensuring privacy compliance with regulations such as HIPAA, GDPR etc, and protection against unauthorized access during image transmission. This approach adapts seamlessly across imaging modalities, offering an adaptable solution for enhanced security in digital healthcare.

Results aim to establish a reliable methodology for heightened patient data security through covert embedding, Leading the way for stronger privacy protection in medical imaging. Proposed solutions include advancements in data hiding, security, robustness testing and adaptive steganographic algorithms. A systematic adaptation of existing techniques tailored to medical imaging requirements, with iterative refinement based on empirical results, aims to yield a secure and practical solution for enhancing medical image security through steganography

Index Terms—Steganography, Medical Imaging

1. INTRODUCTION

The COVID-19 epidemic has brought to light the importance of multimedia technologies and the Internet. As we communicate private information online in our daily lives such as bank account details, credit card numbers, health information, or family photos we are growing increasingly indifferent to internet communication tools.

These days, disclosing such private or business information online carries a considerable risk [3]. Hence securing sensitive medical data of the health care institutions, such as patient information and diagnostic images, poses a significant challenge in the digital age [5]. Traditional encryption methods can raise red flags for unauthorized access attempts [15]. Additionally, medical images often contain crucial diagnostic information that needs to be preserved during security protocols [9].

From a social standpoint, identity theft resulting from healthcare data theft not only affects credit scores and insurance eligibility but also undermines faith in healthcare organisations, discouraging people from obtaining essential medical care [11]. When seen from a humanitarian perspective,

concerns over data breaches pose a threat to international health initiatives as well as impeding study participation [13].

For prompt treatment and precise diagnosis, the secure transmission of medical pictures becomes essential in resource-constrained environments and during humanitarian situations [7]. Breach scenarios in these circumstances might prevent impoverished communities from receiving necessary medical care, which would hinder global health initiatives [14].

Furthermore, The financial consequences of data breaches put a financial burden on healthcare organisations, increasing patient costs, and necessitating costly compliance procedures and security upgrades [6]. These outcomes underscore the far-reaching effects of compromised healthcare data on both individuals and the broader healthcare ecosystem [8]. This project explores using steganography, the art of hiding information within another medium, to improve medical imaging quality, security and privacy [2]. This approach aims to:

Implementing steganographic techniques like LSB [16], CNR [4], MSE [9], PSNR [9], SVM [12] and IWT [7], compliance with HIPAA standards can significantly enhance data security and privacy in the healthcare sector [11]. This approach enables individuals to confidently share sensitive information, ensuring timely access to essential care [13]. Particularly crucial in resource-limited settings, steganography facilitates the secure transmission of medical images, supporting vital diagnoses and treatments for vulnerable populations [14]. This not only minimizes data breaches but also contributes to more efficient and cost-effective healthcare by reducing associated costs [6]. In addition, by fostering trust and security, steganography promotes preventive care, early diagnosis, and overall better health outcomes for individuals, thus improving the overall well-being of the community [8].

2. METHODOLOGY

2.1 Steganography

Steganography is the art of concealing secret information embedded in the carriers, such as images, audio files, or text. It has emerged as a crucial tool in the realm of digital security and privacy [15]. While encryption scrambles data to render it unreadable, steganography operates by hiding the existence

of the information itself. One particularly promising area of application is in the domain of medical imaging [2].

Moreover, the implementation of steganography in medical imaging carries significant consequences for cybersecurity in the healthcare industry. Enhancing the security of medical pictures helps healthcare organisations defend patient confidentiality and trust by reducing the likelihood of data breaches, unauthorised access, and manipulation [11].

2.2 Least Significant Bit

The Least Significant Bit (LSB) constitutes a fundamental concept within the realm of digital information representation, primarily in computer science and digital signal processing [16]. With the LSB in the rightmost place in a binary sequence, binary numeral systems use binary digits or bits to convey information. It is significant because, with a weight of 2^0 or 1, it represents the lowest numerical value in the binary format. When binary numbers are read from right to left, the least significant bit is the LSB, and the bits that follow it to the left get more and more significant by powers of 2.

The LSB is essential for digital data processing, communication, and storage in real-world applications [15]. Changes to the low-level bit (LSB) of certain pixels or samples are very useful in picture, audio, and video files for steganography and even simple data encoding. The LSB is also essential in the fields of encryption and cryptography, since it may be used to conceal or protect data, facilitating the safe transfer of information [2].

2.3 Contrast to Noise Ratio

The Contrast-to-Noise Ratio (CNR) is a quantitative measure widely employed in imaging and signal processing to assess the distinguishability of objects or features against the background noise within a given system [4]. This metric is particularly crucial in fields such as medical imaging, where the clarity of anatomical structures is paramount. CNR is determined by comparing the noise, which is an indication of pixel values random fluctuations or variability, with the contrast, which shows the intensity difference between two neighbouring regions of interest [9].

2.4 Mean Square Error

In the realm of statistics and related fields like signal processing, the mean squared error (MSE) emerges as a crucial metric for gauging the quality of an estimator [9]. It quantifies the average squared discrepancy between the actual values and the corresponding estimated values.

In simpler terms, it reflects how close, on average, the estimations are to the true values. The MSE for an estimator is calculated as the expected value of the squared difference between the estimated value and the true value. A lower MSE signifies a better estimator, indicating that the estimations, on average, are closer to the true values. Conversely, a higher MSE implies a poorer estimator, suggesting the estimations are on average, further away from the true values [8].

2.5 Peak signal-to-noise ratio

Peak signal-to-noise ratio (PSNR) is a commonly used statistic in signal processing and image/video compression to assess reconstruction quality [9]. It explores the power dynamics between the unaltered original signal and the undesired noise that tampers with its accurate representation. Because of the frequently wide dynamic range of signals, PSNR is represented in decibels (dB), but a logarithmic scale is more suited for understanding.

The ratio of the greatest potential signal power which is frequently determined by the bit depth to the power of the corrupting noise is used to compute PSNR. Greater PSNR values correspond to a more robust signal relative to noise, suggesting a more accurate reconstruction. Lower PSNR values, on the other hand, indicate a poorer reconstruction as they indicate a weaker signal in comparison to the noise. While a higher PSNR generally suggests a better reconstruction [12].

2.6 Support Vector Machine

The Support Vector Machine (SVM) is a supervised classification method derived from statistical learning theory [7]. SVM aims to find an optimal hyperplane in a high-dimensional space that effectively separates different classes within the dataset. It's often used to yield good classification results from complex and noisy data. SVM separates the classes with a decision surface that maximizes the margin between the classes.

The key innovation of SVM lies in its ability to maximize the margin, which is the distance between the hyperplane and the nearest data points from each class. This not only enhances the model's generalization performance but also improves its resilience to outliers. SVM is particularly well-suited for scenarios where the data is not linearly separable [12].

One of the distinctive features of SVM is its structural risk minimization approach, which focuses on minimizing both empirical error and model complexity simultaneously. The algorithm's optimization process involves solving a quadratic programming problem, ensuring robustness and efficiency in finding the optimal hyperplane. SVM has proven to be effective in various domains, such as image classification, text categorization, and bioinformatics. This emphasis on maximizing the margin inherently promotes strong generalization, making SVMs a valuable tool for various real-world applications [8].

2.7 Integer Wavelet Transform

The Integer Wavelet Transform (IWT), is a signal-processing method whose operations are limited to integers [7]. It is especially intended for the analysis and manipulation of signals represented by numbers. Because of this, applications requiring lossless data representation or those with limited computing resources are especially well-suited for the IWT.

The IWT's main concept is to use wavelet functions to break down a signal into its most basic parts [5]. These wavelet

functions allow for a simultaneous study in the frequency and time domains, which offers important insights into the properties of the signal at various scales. At every stage of decomposition, the IWT produces integer coefficients that indicate the contribution of the signal. The integrity of the signal may then be preserved by using these coefficients for other applications including feature extraction, data compression, and denoising [2].

3. FORMAL EXPERIMENT

3.1 Research Goal

In this study, we aim to assess the potential impact on medical imaging by employing steganography, a method that conceals information within other data [15]. Our objective is to examine whether the quality and data privacy of medical images are affected when additional information is embedded using steganographic techniques [2].

The "Formal Experiment Method" is used in this study on medical imaging to carefully control and manipulate factors like image quality and data privacy. This helps ensure reliable and objective results by using random assignment, procedures, and standardized conditions [11]. This method allows researchers to draw clear conclusions about how changes in image quality and data privacy impact medical imaging outcomes, providing valuable insights for improving healthcare practices [8].

3.2 Hypotheses

- **Null Hypothesis (H_0):** There is no significant impact of steganography on the quality of pictures and data privacy in medical imaging.
- **Alternative Hypothesis (H_1):** There is significant impact of steganography on the quality of pictures and data privacy in medical imaging.

3.3 Who will participate?

It is a Human-Computer Interaction(HCI) that involves various participants such as:

- **HIPAA** (Health Insurance Portability and Accountability Act) [11]
- **GDPR** (General Data Protection Regulation) [13]
- **ISO** (International Organization for Standardization) [10]
- **NIST** (National Institute of Standards and Technology) [8]
- Digital Forensic Lab [14]
- Cybersecurity Experts [12]

3.3 What will be measured?

(i) *Image Quality Parameters::*

TABLE I

S.No	Parameter	Metric
1	Contrast	Contrast-to-Noise Ratio (CNR)
2	Resolution	Mean Squared Error (MSE)
3	Noise	Peak Signal-to-Noise Ratio (PSNR)

(ii) *Data Privacy Parameters:*

TABLE II

S.No	Parameter	Metric
1	Data Hiding and Security	LSB (OpenStego Tool), SVM, IWT
2	Compliance with Privacy Regulations	HIPAA, GDPR, CCPA

3.4 How will it be measured?

(i) Image quality parameters will be assessed using the above-mentioned metrics and subjective ratings.

(A) *The contrast of the image can be measured using CNR:*

$$\text{CNR} = \frac{\Delta S}{N}$$

Where: - ΔS is the difference in signal intensity between two regions of interest (e.g., between a target structure and its surroundings). - N refers to the noise, which is the unwanted or random variations in the pixel values of the image.

In medical imaging, such as X-ray, MRI, or CT scans, CNR is often used to evaluate the visibility of structures within the image. Enhancing image contrast can improve the concealment of medical images, while reducing contrast may reveal the hidden content. However, a higher Contrast-to-Noise Ratio (CNR) is considered better because it indicates a stronger signal relative to the background noise, leading to better image visibility [4].

(B) The resolution of the image can be assessed by measuring the Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_i - K_i)^2$$

Where:

- N is the total number of pixels in the image.
- I_i is the intensity of the i -th pixel in the original image.
- K_i is the intensity of the i -th pixel in the reconstructed or compressed image [9].

MSE measures the average squared difference between the pixel values of the original and the steganographic images. A lower MSE value is considered better as it signifies smaller differences between the original and reconstructed images [8].

(C) The noise in an image can be assessed by measuring its Peak Signal-to-Noise Ratio (PSNR):

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{\text{MSE}} \right)$$

Where:

- MAX is the maximum possible pixel value of the image (e.g., 255 for an 8-bit image).
- MSE is the Mean Squared Error, as calculated earlier.

PSNR is expressed in decibels (dB), and higher PSNR values indicate better image quality. The signal represents the original image, while the noise represents any distortions or errors that occur during compression.

Original Cover Image	Stego Image	Performance Metrics
		MSE = 0.0302 PSNR = 63.3309 dB
		MSE = 0.0063 PSNR = 70.1539 dB
		MSE = 0.0215 PSNR = 64.8082 dB
		MSE = 0.0242 PSNR = 64.2903 dB

Fig. 1. Comparison between Original and Stego Image

As evident in this comparison between the original image and the stego image, a lower Mean Squared Error (MSE) and higher Peak Signal-to-Noise Ratio (PSNR) values suggest that the secret message is nearly imperceptible, with minimal observable changes in both images, making them appear virtually identical [8].

The existence of a secret message is extremely low almost non-existent, due to the low value of the MSE and the high value of the PSNR [8].

(ii) Data Privacy parameters will be assessed using the following techniques:

(A) Data Hiding process can be done by following:

1) **Least Significant Bit (LSB):**

The LSB method works by replacing the rightmost bit in the binary representation of the image. For a given pixel, this can be written as:

$$LSB = b[0]$$

Where $b[0]$ represents the rightmost bit in the binary representation of a number [16].

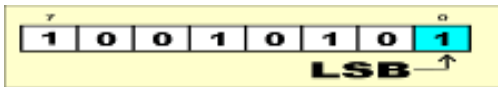


Fig. 2. LSB position in an 8-bit sequence for data hiding

The general value used for LSB replacement in steganography is small to avoid noticeable changes in the images. In the LSB technique, the secret message is embedded in the LSB of the medical image's pixels through encryption. The medical image is first transformed into bytes, with each byte's LSB set to zero. The secret message is then encrypted and transformed into a series of bits that need to be placed into the medical image's vacant LSB spaces. Finally, a stego-image is generated that looks identical to the original medical image but contains the secret message by replacing the bits in the LSB with those of the encrypted secret message [15].



Fig. 3. Medical Report before steganography



Fig. 4. Medical Report after steganography

The figures show almost no visible changes, suggesting that the LSB method lets you hide data without making any noticeable differences [15].

2) **Another approach is by using SVM and IWT:**

SVM and IWT provide an advanced strategy for concealing data and facilitating secure information transmission [7].

SVM is a new class of supervised machine learning technique that works well for classification issues and may also be applied to regression [12]. Each data item in this classification technique is represented as a point in an n-dimensional feature space of items with a given coordinate value. Next, a classification is carried out using the best hyperplane, as shown in Figure, to separate the two classes. The data points close to the dividing hyperplane are referred to as support vectors. In this case, the margin denotes the maximum width of two parallel lines to the hyperplane, which is devoid of any data points inside the parallel slabs. Points of support vectors are located along the slab's edge. Squares represent the data points in the positive category, while circles represent the data points in the negative category [7]. A key step in embedding hidden information in non-essential areas of medical imaging is categorising ROI and NROI regions. One way to get around it is to use SVM as a classifier, which keeps the diagnostic portion of the picture from being distorted [12]. Thus, this kind of categorization approach can improve upon the schematic steganographic process [15].

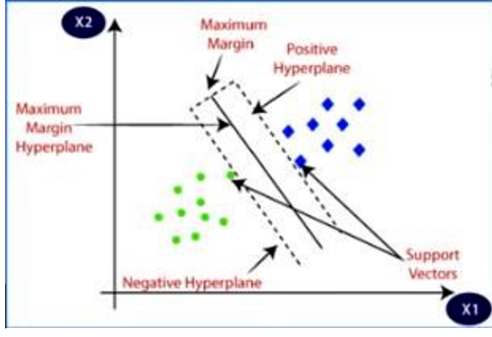


Fig. 5. Diagrammatic representation of SVM hyperplane classification [19]

Integer Wavelet Transform (IWT) is frequently utilised for image processing, steganography, and compression. Its ability to display several resolutions makes various image details visible. The data is divided into discrete frequency components using the **Wavelet function** [16].

IWT's ability to study a signal's discontinuities and abrupt spikes demonstrates its superiority over conventional transform techniques [8]. Two-dimensional IWT at a single level breaks down the signal or image into four nonoverlapping bands called LL, LH, HL, and HH bands. LL stands for approximation coefficients, and LH, HL, and HH stand for detail orientations, such as the horizontal, vertical, and diagonal components of the two-dimensional signal, respectively. There are four further sub-bands that may be identified from the first level's approximation coefficient (LL): LL, LH, HL, and HH. You can keep doing this until you reach the required amount of levels. The original picture is constructed by reassembling the sub-bands using inverse image wavelet transform, as illustrated in Fig. 6.

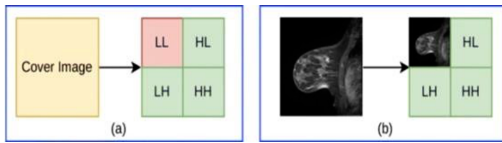


Fig. 6. IWT decomposition of an image [19]

Three subsections have been created for the proposed scheme:

I. Image Classification

In the proposed model's classification stage, SVM is employed to categorize pixels in the medical host image into Regions of Interest (ROI) and Non-ROI (NROI) for embedding [12]. The SVM model is trained using labelled predictor data, with various kernel functions available to enhance predictive precision by adjusting parameters. Post-training, cross-validation is performed on the test dataset. Misclassifications between ROI and NROI are minimized by managing SVM parameters such as the weight vector and bias. Accurate ROI identification is critical for diagnosis, making misclassifications problematic [7].

Figure 5 depicts the proposed model, with the SVM algorithm trained using an MRI dataset. The SVM model predicts ROI and NROI regions in the host medical image. Pixels in NROI regions are chosen for secret data embedding without impacting ROI pixels. Refer to Section 2.2 for the detailed embedding procedure.



Fig. 7. Two phase diagram of steganographic scheme [19]

II. Information Embedding

The proposed embedding scheme begins with a color cover image of size $M \times N$, where the regions of interest (ROI) and non-ROI (NROI) are detected using Support Vector Machine (SVM), yielding pixel positions stored in arrays Arr_{rpos} and Arr_{nrpos} , and pixel values stored in arrays Arr_{rpx} and Arr_{nrpx} [12]. The focus is on the NROI part for embedding. The cover image is split into its RGB components, and each component is subjected to an Inverse Wavelet Transform (IWT) to obtain the coefficient matrices IWT_R , IWT_G , and IWT_B .

The embedding process starts with the red component IWT_R , using a shared secret key x to generate a 512-bit string from x via SHA-512. A stego bit stream B_w is derived from the logo W . Using a circular array and x , stego information is embedded in IWT_R by rotating the array clockwise ($x = 0$) or anticlockwise ($x = 1$). Four stego bits are embedded within nine pixels for each operation [16].

This process is repeated for IWT_R and IWT_B , embedding all secret bits within the NROI of the cover image. Inverse IWT is then applied to the three color components, and the R , G , and B components are merged to produce the stego image (SI).

III. Information Extraction

The secret bit extraction process begins with the SI of size $M \times N$, followed by SVM-based separation of the ROI and NROI parts, resulting in pixel positions stored in Arr'_{rpos} and Arr'_{nrpos} , and pixel values in separate arrays Arr'_{rpx} and Arr'_{nrpx} . The focus is on the NROI part containing hidden information bits [12].

The image is then split into R, G, and B color components, with Inverse Wavelet Transform (IWT), applied to each component to obtain coefficient matrices IWT'_R , IWT'_G , and IWT'_B . Extraction starts with the red component IWT'_R , using the shared secret key X to generate a 512-bit string through SHA-512. This key, κ , determines the rotation direction of the circular array, consistent with the embedding process [16].

The circular array is then employed to extract stego bits from IWT'_R and store them within B'_w , using X . Four secret bits are extracted from nine pixels in each operation. This process is repeated for the green IWT'_G and blue IWT'_B color coefficient blocks [9].

Following that, inverse IWT is applied to the coefficient blocks of all three color components to obtain pixel values in image format. The R , G , and B components are merged to generate the cover image (I), and the extracted secret bits are appended to reconstruct the original logo image [7].

(B) Compliance with Privacy Regulations:

It ensures that steganographic techniques comply with healthcare privacy laws, such as the Health Insurance Portability and Accountability Act (**HIPAA**) in the United States, **GDPR** and **CCPA** [11], [13].

Medical imaging data is protected by HIPAA, the Health Insurance Portability and Accountability Act, which also guarantees data security and privacy. Following HIPAA regulations is essential when employing steganography in medical imaging. The Security Rule requires data integrity and access controls, the Privacy Rule regulates patient confidentiality, and risk assessments must take steganography into account to secure electronically protected health information (ePHI). Data integrity and patient privacy are protected by compliance [8].

Standards set by organisations like the National Institute of Standards and Technology (**NIST**) provide frameworks for securing information systems, including medical imaging systems [14].

The use of advanced technologies (such as Artificial Intelligence), automation in implementing security updates and continuous monitoring can benefit medical imaging systems from dynamic security that adapts to threats [12].

3.5 Data Collection Procedure

The process used for data collection comprises the following steps:

- **Participant Recruitment and Consent:** In this step, medical imaging experts are recruited and provide their consent to participate in the study [11].
- **Random Assignment and Blinding:** Participants are randomly assigned to different groups, and blinding techniques are employed to minimize bias [7].
- **Introduction and Image Presentation:** A brief introduction to the study is given, followed by the presentation of medical images, both with and without steganography information [8].
- **Image Quality and Privacy Assessment:** Participants are instructed to evaluate the image quality using standardized techniques. Similarly, privacy risks are assessed by the participants [12].
- **Data Recording:** Quantitative data is recorded based on image quality and privacy risk evaluations [9].
- **Debriefing:** After the assessments, participants are debriefed to clarify any questions and provide further context as needed [14].

- **Data Analysis:** Statistical analysis is performed on the image quality metrics (e.g., LSB, CNR) and privacy risk assessments to draw meaningful conclusions [16].
- **Result Interpretation and Reporting:** The results are analyzed to determine the insights that can be gained regarding privacy and medical images [15].

3.6 Data Analysis Strategy

The data analysis process for the proposed system follows a structured approach:

- **Identifying Patient Information:** The patient's information requiring protection is identified and converted into a secure, compressed format [13].
- **Choosing Steganographic Techniques:** Steganographic techniques such as Least LSB, Discrete Wavelet Transform (DWT), and Support Vector Machine (SVM) are selected based on the medical imaging data [16].
- **Enhancing Data Privacy:** Metrics related to LSB, DWT, and SVM are implemented to improve data privacy in medical images [9].
- **System Integration:** Steganographic methods are integrated with medical imaging systems, ensuring compatibility with formats like DICOM (Digital Imaging and Communications in Medicine) [14].
- **Image Quality and Privacy Assessment:** Image quality and privacy risks are assessed using distinct metrics for accurate evaluation [12].
- **Validation and Testing:** Rigorous validation and testing are conducted to ensure system robustness [8].
- **Documentation and Reporting:** Results are documented comprehensively to summarize findings and outcomes.

4. LIMITATIONS AND RISK MANAGEMENT

1. Potential Validity Threats

- **Data quality and integrity:** Medical images used for steganography need to be of high quality and free from artifacts. Poor image quality could affect the embedding process and lead to errors in extracting hidden data [15].
- **Participant related biases:** The Human-Computer Interaction based evaluation relies on expert judgments of image quality and data privacy. Differences in participant's experience with steganography, cybersecurity and medical imaging may introduce bias, as more experienced experts can detect artifacts or privacy risks that others may miss. This threatens construct and internal validity. To reduce this risk, participants are grouped by expertise, receive identical instructions, and use standardized rating scales during the assessment [8], [9].
- **Security vulnerabilities:** Steganographic techniques themselves might have vulnerabilities that could allow unauthorized access to hidden data. Robust algorithms and secure key management are essential [9], [12].

2. Mitigating Risks

a) Operational Risks:

Risk	Description	Mitigation
Unauthorized access to stego images	Unauthorized access to stego-images risks exposing sensitive hidden data, like medical records or confidential information, due to steganography's covert nature and potential for human error [9], [11].	Implementing stringent access restrictions for storage systems and databases. Compliance with HIPAA regulations ensures data privacy and introducing encryption for further security [13].
Human error during embedding/extracting	Human error in steganography can corrupt hidden data, prevent extraction, and lead to unintended leaks. Mitigating this by using standardized procedures, automated tools, and proper training [8], [12].	Implementing training programs for personnel involved in the steganography process. Incorporating automated tools and appropriate validation procedures [14].

b) Technological and Methodological Risks:

Risk	Description	Mitigation
Software Compatibility Issues	Different file formats and software versions can lead to data errors. Security risks may arise from integrating external tools [14].	Regularly updating and testing software compatibility, having alternative software solutions, and ensuring a backup plan in case of unforeseen issues [12].
Hardware Failures	Hardware flaws could allow security to be breached, revealing sensitive patients' data [8].	Regular maintenance and monitoring of hardware, having backup equipment, and establishing a rapid response plan for hardware failures [9].

5. CONCLUSION

This study rigorously examines the implementation of sophisticated steganographic methodologies for securing sensitive patient information within medical imaging frameworks. By leveraging techniques such as Least Significant Bit (LSB) substitution, Support Vector Machine (SVM) classification, and Integer Wavelet Transform (IWT) decomposition, the research substantiates the potential for confidential data embedding that preserves diagnostic integrity. Analytical evaluation employing metrics such as Contrast-to-Noise Ratio (CNR), Mean Squared Error (MSE), and Peak Signal-to-Noise Ratio (PSNR) demonstrates the exceptional quality of stego-images, affirming the imperceptibility of embedded data and its negligible impact on clinical functionality [9].

The proposed framework is intricately aligned with international data protection standards, including HIPAA and GDPR, emphasizing its applicability in mitigating contemporary challenges of digital healthcare security. The methodology not only fortifies patient confidentiality but also facilitates secure data transmission in resource-constrained and emergency medical contexts. By enhancing compliance and trust, this approach offers a scalable and robust pathway for addressing risks associated with data breaches and unauthorized access [13].

Nevertheless, the investigation identifies critical challenges, such as human-induced errors in embedding and extraction processes, algorithmic susceptibilities, and compatibility constraints with hardware systems. Addressing these challenges necessitates future efforts focused on augmenting algorithmic resilience, automating operational workflows, and engineering adaptive methodologies compatible with diverse imaging modalities. Furthermore, the incorporation of machine

learning and artificial intelligence can refine the efficiency and scalability of steganographic techniques, broadening their applicability within dynamic medical environments [8].

In conclusion, this research establishes a pivotal foundation for advancing secure, efficient, and privacy-compliant data management strategies in healthcare. By integrating innovative steganographic approaches, it contributes to the development of resilient, patient-focused digital ecosystems that uphold the integrity and security of sensitive medical information [16].

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